

Md Karimul Islam

AI / ML Engineer · Computer Vision · LLM & Prompt Engineering · XAI

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EXPERIENCE

Prompt Engineering Intern · Oleyn Jan – Apr 2026
PTE Ltd. (Oley.ai) — Remote

- Engineered structured prompts for production LLM pipelines across 3+ task categories using chain-of-thought and role-based strategies, cutting iteration cycles measurably.
- Benchmarked model outputs on reliability and quality metrics; all milestones certified by Co-Founder & CEO (Future Leaders Program, Batch 1).

Founder & CEO · GeoChain — Rajshahi Apr – Sep 2025

- Built a blockchain-ready property marketplace with verification workflows; secured 2nd UIHP grant — two-time grantee in 12 months.

Founder & CEO · VetConnect — Rajshahi Oct 2024 – Mar 2025

- Launched Bangladesh's first digital veterinary platform; delivered GPS discovery, telemedicine, and e-commerce prototype in 6 months; UIHP-funded.

RESEARCH & PUBLICATIONS

MEFNet — Hybrid CNN for Waste Classification + XAI · First Author · IEEE PECCII 2025–2026

- MobileNetV3 + EfficientNetB0 fusion; **97.71% accuracy** across 12 categories; Grad-CAM interpretability — outperformed ResNet50V2, EfficientNetV2, ConvNeXt-Tiny.

FusionNet — Deepfake Detection Framework · First Author · VIJIR Vol. 1(2) 2025 · UCICS 2025

- Hybrid DenseNet121/169; **91.22% accuracy, AUC 0.98, F1 92.56%**; adversarial robustness tested against FGSM attacks.

LitePhospho — XAI CNN for Phosphorylation Prediction · Undergraduate Thesis · Ongoing

- Lightweight 1D-CNN with Integrated Gradients XAI; ONNX export for low-resource deployment; homology-aware benchmarking.

Enhanced MobileNetV1 for Autism Detection · 7th Author · IEEE QPAIN 2025

- Co-authored early autism detection model using enhanced MobileNetV1 architecture.

EDUCATION

B.Sc. CSE · Varendra University, Rajshahi Sep 2022 – Aug 2026

- CGPA **3.90 / 4.00** · Merit Scholarship (2022–Present)
- Hult Prize Campus Team Leader (2026)

SKILLS

ML / DL: PyTorch · TensorFlow · Keras · Scikit-Learn

Architectures: CNNs, DenseNet, ResNet, MobileNet, EfficientNet, 1D-CNN

CV / XAI: OpenCV · MediaPipe · Grad-CAM · Integrated Gradients · LIME · ONNX

LLM: Prompt Design · CoT · Role-based · LLM Evaluation

Programming: Python · C++ · C · SQL

Tools: Docker · Linux · Git · Figma · Tesseract OCR · Arduino

PROJECTS

Agentic Research Assistant | RAG · LLM Agents 2026
· FAISS · ONNX · [Live Demo](#)

- LLM agent answering across research papers with page-level citations; scores its own answers for faithfulness & relevance (LLM-as-judge).

Multimodal Visual Analyst | Vision-Language 2026
Model · Llama 4 · [Live Demo](#)

- Upload an image to ask questions or extract a structured JSON report (objects, on-image text, insights) via a VLM.

Audio Intelligence | Whisper · LLM · Streamlit · 2026
[Live Demo](#)

- Transcribes meetings/lectures, then generates a structured report (summary, action items) with Q&A over the transcript.

Churn Prediction — MLOps | FastAPI · scikit-learn · Docker · CI/CD · [Live Demo](#)

- Self-trained model served via FastAPI with validation, a prediction form, Prometheus metrics, a live dashboard, Docker & CI.

Driver Drowsiness Monitor | Python · MediaPipe 2023
· OpenCV

- Real-time fatigue detection using facial landmark tracking — blink-rate and head-pose metrics with low-latency inference for driver safety alerts.

Secure ALPR System | Python · OpenCV · Tesseract OCR · SQLite 2024

- Real-time automatic license plate recognition with encrypted SQLite logging and firewall-based access control for secure deployment.

MyCashApp — Banking Simulator | C++ · File I/O 2023

- C++ banking simulator with persistent storage, transfers, bill payments, and account reporting.

CERTIFICATIONS

Google AI Agents Intensive (5-Day) · ML with Python
Data Visualization with Python · Google Analytics

Part-time / remote now · Full-time from Aug 2026